

Cost vs Performance Trade-offs (200-word Reflection)

The central cost vs performance tension in this project manifests most clearly in streaming infrastructure choices. Running Spark Structured Streaming on a continuously active Databricks cluster provides sub-minute data freshness but incurs constant DBU charges. A scheduled micro-batch approach (every 15 minutes) reduces cluster costs by roughly 90% while accepting slightly stale analytics — an acceptable trade-off for non-real-time dashboards.

Kafka retention is another cost lever: 7-day retention enables full pipeline replay for debugging but multiplies Confluent Cloud storage costs with data volume. Reducing retention to 24 hours would be appropriate once the pipeline reaches production stability.

Delta OPTIMIZE and ZORDER operations improve query performance on Gold layer tables by 3-5x but require additional compute time. Scheduling these during off-peak hours balances cost with analytical speed. Overall, a hybrid strategy — continuous streaming only for mission-critical real-time KPIs, batch for everything else — provides the optimal cost-to-latency ratio for a retail analytics platform of this scale.